SLIIT
FACULTY OF COMPUTING

Final Presentation

AI Powered Smart Orchid Care System Using Multi-Modal Machine Learning



Group R26-SE-018

Software Engineering | Faculty of Computing, SLIIT

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Project Team

Supervision



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Group members



Wickramasinghe D.P

IT22065230

Hybrid Pollination and Compatibility Analysis



Rathnamalala D.M.J.P

IT22064936

Disease Detection and Treatment Recommendation



Vithanage P.V.P.M

IT22062642

Smart Watering and Automated Fertilization



Nimla Samadinee L.D

IT22083500

Growth Stage Recognition and Bloom Prediction

PP2 Panel Comments and Responses (1 of 2)

Member	PP2 panel comment	How it is addressed
IT22065230 Wickramasinghe D.P Hybrid Pollination and Compatibility Analysis	Need to clearly identify what comparison mechanisms are used, and didn't show the plant detection flow or how it happened. It needs to show evaluation metrics results during the presentation or demonstration.	**Comparison mechanisms.** Four classifiers compared under the same grouped cross validation, and XGBoost kept. Each parent is judged against its role, with a stricter standard for the pod parent. **Plant detection flow.** Photo → input check → traits read from the photo → readiness model → verdict. Shown on slide 7 and demonstrated live. The cross check is a separate path — it needs two parent names, not a photo **Evaluation metrics.** Accuracy, balanced accuracy, precision, recall and F1 for every model, scored by plant. Readiness 71.7 % against a 63.9 % baseline. The input check refuses 92 % of non plant photos (455 held out images).
IT22064936 Rathnamalala D.M.J.P Disease Detection and Treatment Recommendation	She needs to show evaluation metrics results during the presentation or demonstration. When identifying diseases, what would happen if humans make mistakes when uploading images (like not relevant images, etc.), need to be a mechanism to identify those before going to the model for the decision. Model accuracy is 50 % with the 2 epochs and haven't tried to improve the accuracy or check with different evaluation metrics.	**Irrelevant images.** A stage 0 check stops photos unlike the training images before the disease model runs: 15 of 17 non-orchid photos refused, no orchid photo refused. Below 70 % confidence the answer is "Not sure". **Contributions.** Each contributed photo is checked at upload and approved by an orchid expert from a research institute before it counts. A new disease needs 30 approved photos from 3 growers; retraining is not automatic. **Accuracy.** MobileNetV2, EfficientNetB0 and a CNN trained from scratch were compared. The model trains for 21 epochs, 16 plus 5 of fine tuning, and reaches 80.6 % on 67 held out photos. **Different metrics.** Macro F1 0.78, kappa 0.70, ROC-AUC 0.93, precision, recall and a confusion matrix. Severity within one grade: 90.7 %.

PP2 Panel Comments and Responses (2 of 2)

Member	PP2 panel comment	How it is addressed
IT22062642 Vithanage P.V.P.M Smart Watering and Automated Fertilization	When finding the best locations (pods), should properly identify what kind of data is used and explain that information clearly as a flow when you demonstrate it. And think about the variations in the sensor readings as well. It needs to show evaluation metrics results during the presentation or demonstration.	**Data and flow.** Temperature, humidity and VPD from every node over the calibration days pass through eight steps, from cleaning to choosing the layout. **Sensor variation.** The flow starts with all nodes side by side, so each sensor's offset is measured and removed. A difference smaller than two sensor errors (±0.5 °C) is not treated as real. **Evaluation metrics.** Tested on 14,620 records that the models never saw. Watering time is 8.4 minutes off on average (R^2 0.95). The humidity-drop warning is right 91.5 % of the time (AUC 0.986 and F1 0.759.
IT22083500 Nimla Samadinee L.D Growth Stage Recognition and Bloom Prediction	She needs to show evaluation metrics results during the presentation or demonstration. Try to get the sensor data directly to the system. The answers were not very satisfactory.	**Evaluation metrics.** Both models retrained on 13 October 2026. Growth stage accuracy 94 % with macro F1 0.94 on 1,768 test images. Bloom date error 0.92 days, RMSE 1.41 days, R^2 0.94. **Sensor data.** The bloom prediction screen fills temperature, humidity and light from the sensor feed in Firebase, with a manual override. **Answers.** Every figure comes from the saved evaluation reports, with precision and recall for each stage and a confusion matrix ready to show.

Research Problem

Research question

Vanda orchids are widely grown in tropical regions like Sri Lanka, but cultivation remains a manual, expertise-driven process. Growers face four key challenges that lack an integrated digital solution.

Key Challenges

Hybrid pollination	Parent plants are chosen from experience. Cross records list only successes, so success cannot honestly be predicted from photos alone.
Disease detection	No accessible, unified system to detect, grade severity, and recommend treatments for local conditions.
Smart watering	The exposed aerial roots are easily damaged by reactive watering schedules . Proper care requires precise timing and nutrient balance.
Growth and bloom	Flowering timing is difficult to predict. Research on Vanda flowering physiology is limited , forcing growers to rely on guesswork for harvest and sales planning.



Solution Overview

Shade house

Sensor node in every section

ESP32 with a DHT22 for temperature and humidity and a BH1750 for light

Master controller

Relay board and 12 V pumps for watering and the humidity tray. Confirms every run.

readings →

← commands

Cloud

Firebase Realtime Database

Readings, commands and confirmations, kept separate for every farm

FastAPI backend

The models of all four components and the automation engine, on a VPS

results →

← requests

Phone

Mobile app (React Native)

One app for the admin, operators and viewers of a farm

Push alerts

Firebase Cloud Messaging when the farm needs a person

Four components on the shared backend

C1 Hybrid Pollination and Compatibility Analysis	**C2** Disease Detection and Treatment Recommendation	**C3** Smart Watering and Automated Fertilization	**C4** Growth Stage Recognition and Bloom Prediction

IT22065230 | Wickramasinghe D.P



Hybrid Pollination and Compatibility Analysis

How it works

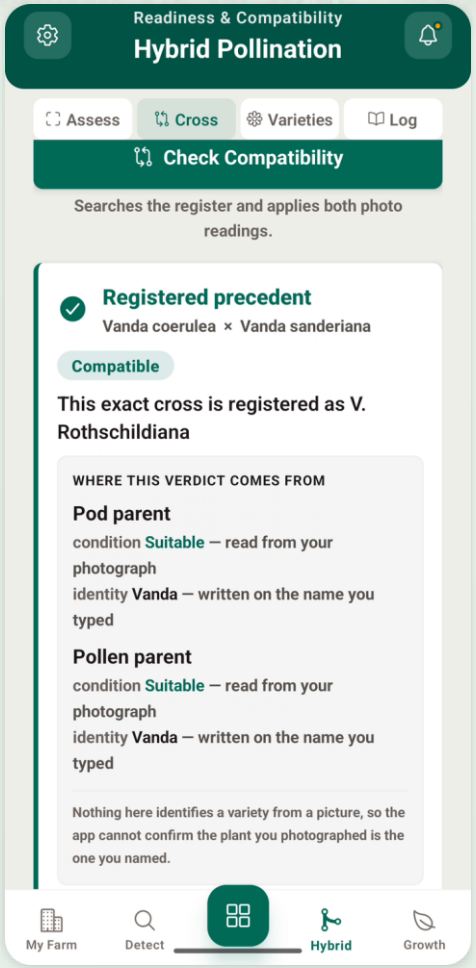
1 Photo	2 Input check	3 Traits	4 Readiness	6 Verdict
of the plant	two stages, non orchids stopped	read from the photo	XGBoost	with its level of evidence

What it does

- Readiness is read from one photograph. 62 colour, shape and texture features, plus 4 recorded plant traits, give 66 features.
- Three level verdict, Suitable, Moderate or Not Suitable, with a stricter standard for the pod parent than the pollen parent.
- The register of 27 crosses holds only crosses that worked,The crossing logbook records the grower's own attempts through 8 stages, including the failures.

Evaluation

Readiness accuracy, cross validated by plant	**71.7 % (baseline 63.9 %)**
Precision, recall and F1 of the readiness model	**0.72, 0.72 and 0.71**
Non orchid plant photos refused, 455 held out images	**92% ROC AUC 0.921**



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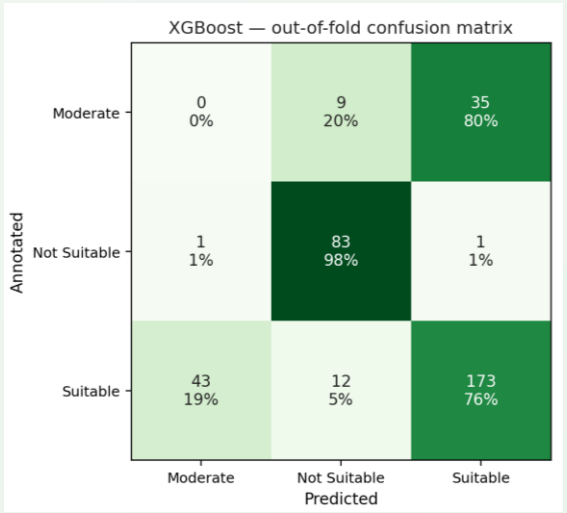


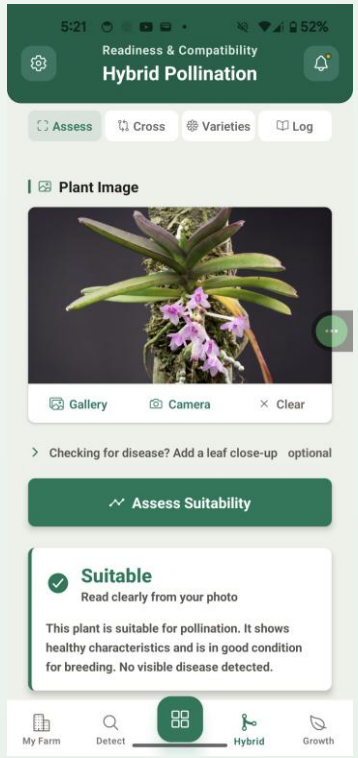
Hybrid Pollination and Compatibility Analysis

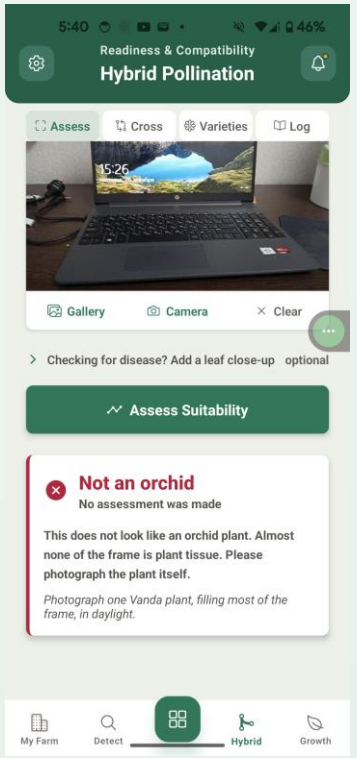
Suitability Models

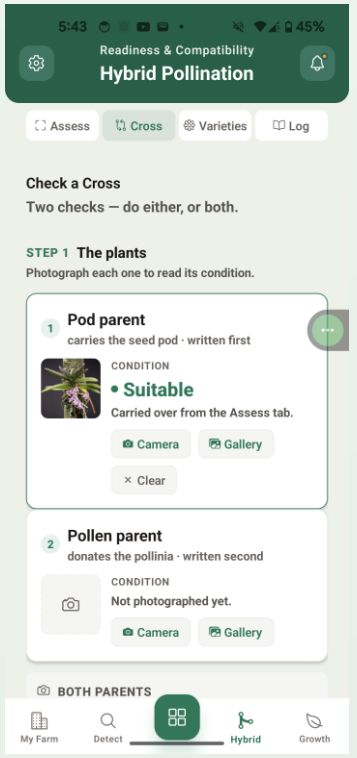
Model	Accuracy	Balanced	F1	vs baseline
XGBoost	**71.7%**	**57.8%**	**0.71**	**+7.8**
Logistic Regression	69.8%	56.3%	0.69	+5.9
Random Forest	52.9%	40.4%	0.52	-10.9
SVM (RBF)	47.9%	40.5%	0.48	-16.0
Majority baseline — always "Suitable"	*63.9%*	—	—	—

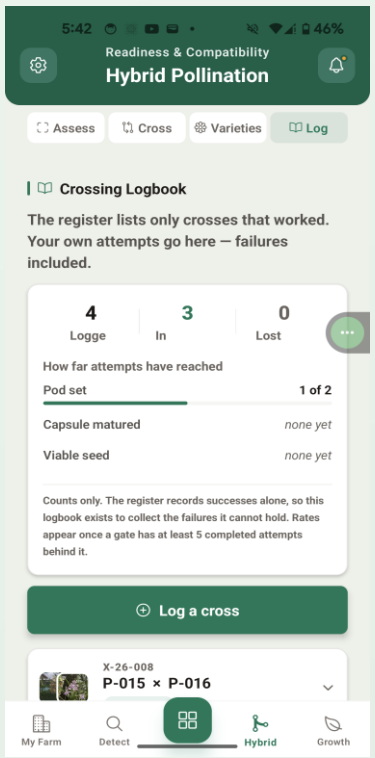
Confusion Matrix











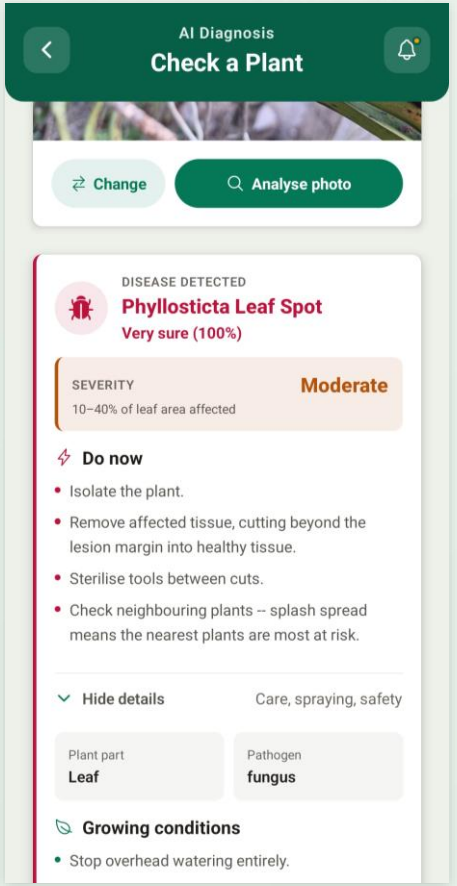
IT22064936 | Rathnamalala D.M.J.P



Disease Detection and Treatment Recommendation

How it works

1 Photo	2 Stage 0 check	3 Disease	4 Severity	5 Treatment
of a leaf in the field	photos unlike training stopped	MobileNetV2, below 70 % unidentified	mild, moderate or severe	for the disease and grade



What it does

- Identifies Black Leaf Spot, Phyllosticta Leaf Spot or a healthy leaf from an ordinary field photo.
- Grades the severity as mild, moderate or severe, and recommends a treatment for that disease and grade.
- MobileNetV2 was compared with EfficientNetB0 and a CNN from scratch (validation macro F1 0.82, 0.83 and 0.60).
- Growers send photos, checked at upload and approved by a research-institute expert. (For diseases not in system)

Evaluation

Test accuracy, 67 held out photos	**80.6 %**
Macro F1 and weighted F1	**0.78 and 0.80**
Severity within one grade (exact grade)	**90.7 % (46.5 %)**

IT22064936 | Rathnamalala D.M.J.P

Disease Detection and Treatment Recommendation



IT22062642 | Vithanage P.V.P.M



Smart Watering and Automated Fertilization

How each section is watered and fed

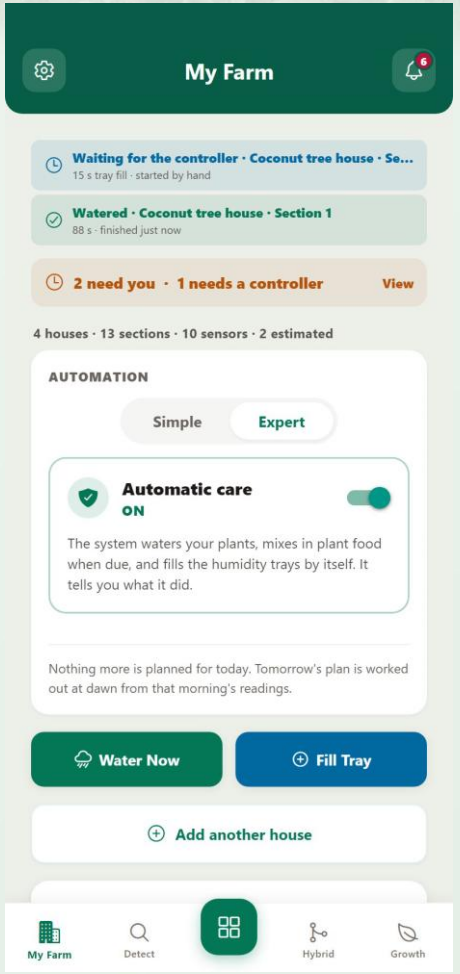
1 Sense	2 VPD	3 Plan	4 Pour	5 Tray	6 Feed	7 Check	8 Alert
T, RH, light each minute	drying power worked out	plan time and duration	master runs the pump	fill, or fill early	plant food by growth stage	monitor all nodes	alarm on the phone

What it does

- Each section gets its own watering time from random forest models trained on 58,436 records of real data.
- One master controller runs the pumps and confirms every watering. The humidity tray is filled before humidity leaves the safe range.
- Fertilizer is scheduled by growth stage. Alarms reach the phone when a sensor, a node or the master stops working.

Evaluation

Watering time error	**8.4 min (R^2 0.95)**
Humidity drop below 60 %	**AUC 0.986, F1 0.759**



Readings shown are demo data

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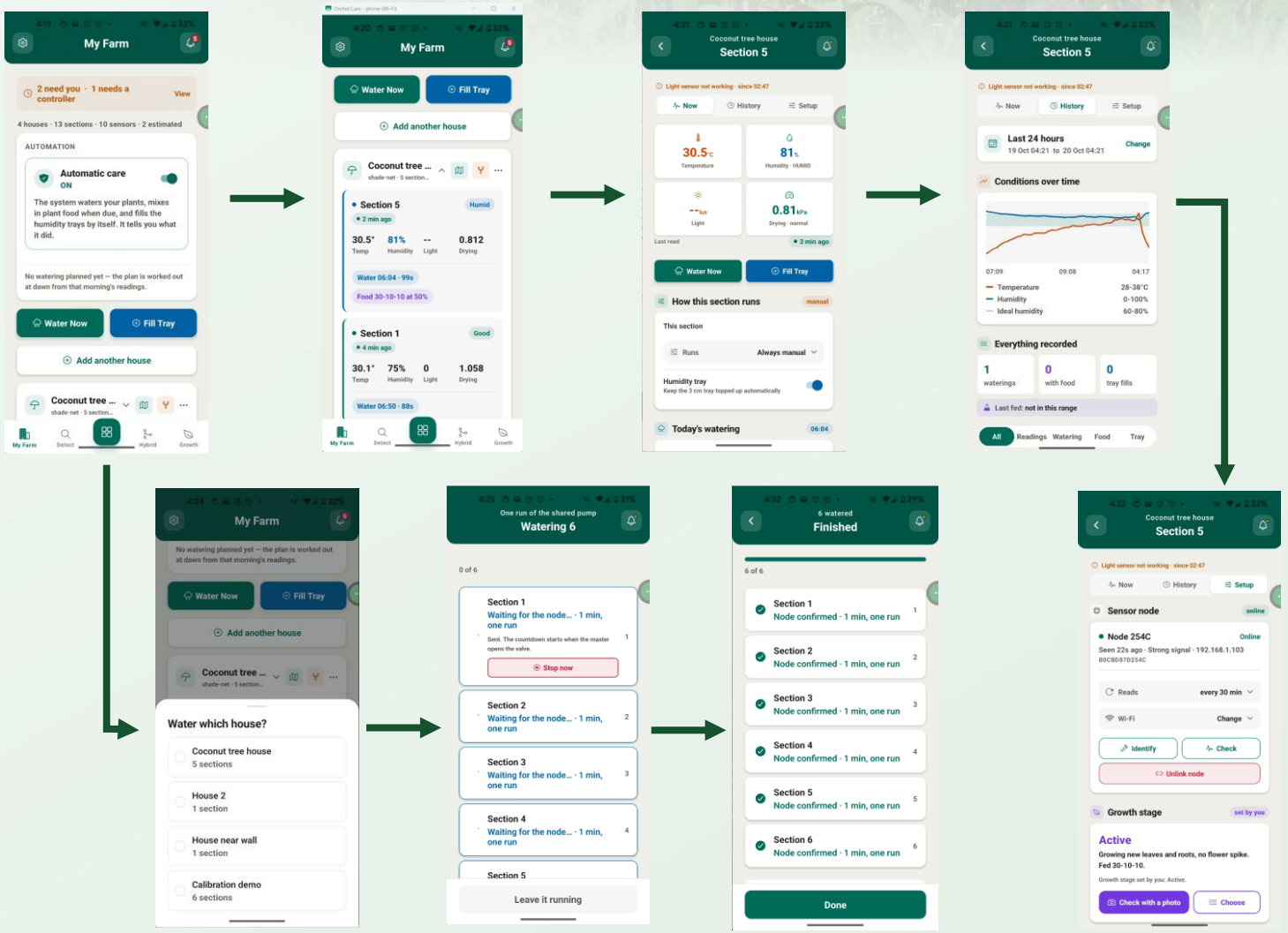
Smart Watering and Automated Fertilization

Watering time

Method	Average error (min)	RMSE (min)	R^2	Within 10 min
Same time every day (the average)	41.03	49.95	-0.008	14.9 %
Linear regression	9.84	12.91	0.933	59.8 %
Decision tree (depth 16)	10.57	13.64	0.925	56.3 %
K nearest neighbours (k = 15)	8.98	11.33	0.948	63.3 %
Gradient boosting (histogram)	8.24	10.39	0.956	67.5 %
Random forest, 200 trees, depth 16	8.42	10.63	0.954	66.4 %
Random forest, 50 trees, depth 16 (in the app)	**8.47**	**10.71**	**0.954**	**66.2 %**

Humidity drop

Method	ROC area	Avg. precision	Precision at 0.70	Recall at 0.70	F1
Always 'no drop' (baseline)	0.500	0.097	0.0 %	0.0 %	0.000
Logistic regression	0.983	0.883	90.7 %	62.1 %	0.738
Decision tree (leaf 25)	0.965	0.849	87.0 %	65.9 %	0.750
Gradient boosting (histogram)	0.986	0.898	89.5 %	68.3 %	0.775
Random forest, 200 trees, leaf 5	0.987	0.905	91.9 %	65.0 %	0.762
Random forest, 50 trees, leaf 120	0.985	0.892	91.5 %	62.1 %	0.740
Random forest, 60 trees, leaf 25 (in the app)	**0.986**	**0.901**	**91.5 %**	**64.8 %**	**0.759**



IT22083500 | Nimla Samadinee L.D



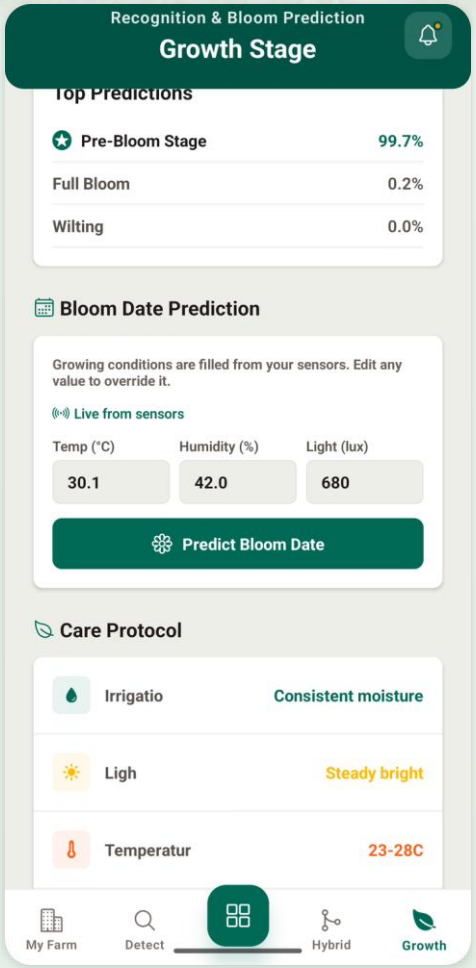
Growth Stage Recognition and Bloom Prediction

How it works

1 Photo	2 Growth stage	3 Conditions	4 Bloom date	5 Care
of the plant	EfficientNet, six stages	temperature, humidity, light	photo and conditions together	protocol for the stage

What it does

- Recognises six growth stages from a photo, including the post flowering stages, wilting and seed formation.
- Predicts the bloom date from the photo with temperature, humidity and light, filled from the sensor feed with a manual override.
- Gives a care protocol for the stage the plant is in.



Evaluation

Growth stage accuracy, 1,768 test images	**94 % (macro F1 0.94)**
Bloom date error, MAE and RMSE (1,839 test images)	**0.92 and 1.41 days**
Bloom date R^2	**0.94**

Test Results

Component	What was tested	Data	Result
C1 Pollination	Readiness verdict, cross validated by plant	357 images of 28 plants	**71.7 % accuracy (baseline 63.9 %)**
C1 Pollination	Input check	455 held out images	**92 % of non plant photos refused**
C2 Disease	Disease classifier	67 held out photos	**80.6 % accuracy, macro F1 0.78**
C2 Disease	Severity grade	43 test photos	**90.7 % within one grade**
C3 Watering	Watering time	58 436 records	**MAE 8.4 min, R^2 0.95**
C3 Watering	Humidity drop, up to 3h ahead	58 436 records	**AUC 0.986**
C4 Growth	Growth stage	1,768 test images	**94 % accuracy, macro F1 0.94**
C4 Growth	Bloom date	1,839 test images	**MAE 0.92 days, R^2 0.94**

Every model result is on data held out from training.

User Review



Reviewer

Nimal Perera







Software Engineering Practices

CI/CD pipeline on GitHub Actions

Push or pull request
Any branch. Each component is built on its own branch.

→

Automatic tests
Backend tests (Python 3.12), app tests (Node 20) and a dependency check

→

Merge to main
Only through a pull request.

→

Automatic deploy
Starts only if the tests passed.

→

Health check
Restarts, checks it started, rolls back if not.

Practice	How we used it
Version control	One GitHub repository, a branch per component, 402 commits and 80 merged pull requests
Automated tests	488 backend tests and 94 app tests. The logic tests run on every push.
Integration tests	Run by hand only, because they send real commands to the farm's pumps
Security	Firebase sign-in, admin, operator and viewer roles checked on the server, secrets kept out of the repository
Separate farms	Each farm is its own tenant, and tests check that one farm cannot reach another farm's data

Cost and Commercialisation

Cost, parts from purchase receipts

Item	Cost
Sensor node, one per section (ESP32, DHT22, BH1750)	**LKR 2,140**
Master controller, one per house (node and relay)	**LKR 2,390**
Server	**LKR 2,292/monthly**
Push alerts (Firebase Cloud Messaging)	**no charge**
Total	**LKR 6,822**

How it earns

Free app for photo checks

Pollination, disease and growth checks need no hardware, so a grower can start without paying

Hardware kit per house

Sensor nodes, master controller and pumps for automatic watering

Subscription per farm as a service

Hosting, alerts, model updates and support. The owner manages operators and viewers.

For Vanda smallholders, nurseries with breeding programmes and new growers, A house with four sections costs about LKR 11,102 in parts, before tubing.

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Thank You

Questions are welcome

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